CSE-4621 MACHINE LEARNING (Set 001101) **Student id:**

Quiz # 3 **Time: 20 min**

1. A neural network has the following: --7
   - Input: 3 neurons
   - Hidden layer: 3 neurons with **ReLU** activation
   - Output layer: 1 neuron with **sigmoid** activation

Given:

$$\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad W^{[1]} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}, \quad b^{[1]} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$W^{[2]} = \begin{bmatrix} 2 & -1 & 1 \end{bmatrix}, \quad b^{[2]} = 0.5$$

Compute the final output (probability) of the network.

2. An RGB image of size **128 × 128 × 3** is passed through the following layers in sequence: --8

   i. **Conv Layer 1**: 32 filters, kernel size = 7×7, stride = 1, padding = 'same'

   ii. **Average Pooling Layer 1**: pool size = 2×2, stride = 2

   iii. **Conv Layer 2**: 64 filters, kernel size = 5×5, stride = 1, padding = 'valid'

   iv. **Average Pooling Layer 2**: pool size = 2×2, stride = 2

(a) Compute the output shape after each layer.

(b) What is the total number of trainable parameters in the above architecture (Conv Layer 1, Pooling Layer 1, Conv Layer 2, and Pooling Layer 2)?

(c) What is Exploding Gradient?